

XII INTERNATIONAL CONFERENCE

Topical PROBLEMS — of — SEMICONDUCTOR PHYSICS



Prykarpattya, Drohobych,
UKRAINE, May 25–28, 2026



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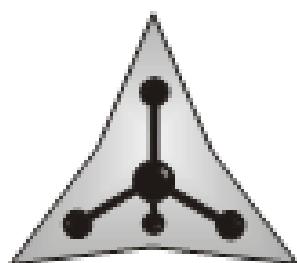
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XII-th International Conference

TOPICAL PROBLEMS OF SEMICONDUCTOR PHYSICS



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Impedance Spectroscopy of Silicon–Polymer Structures Fabricated Using Acoustic Waves

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Silicon–polymer structures offer a promising platform for renewable energy by combining the technological versatility of silicon with the functional flexibility of polymers. Optimizing fabrication to control their performance remains a key objective. This study examines the effect of ultrasound (US) loading during the room-temperature relaxation (RTR) stage on the electrical properties of n-Si–PEDOT:PSS heterojunctions. Primary waves (PW, 2.5 MHz) and radial waves (RW, 500 kHz) were applied to structures with varying polymer thickness controlled by spin-coating (SC) speed.

Frequency-dependent impedance spectra ($2 \cdot 10^{-4}$ –1 MHz) were fitted using a standard equivalent circuit (see Fig) for organic structures [1]. The $R_t || CPE_t$ combination relates to charge transport, whereas $R_r || CPE_r$ accounts for recombination. CPE is defined as $Z_{CPE} = [Q(i\omega)^n]^{-1}$, where n reflects structural inhomogeneity. Key parameters, including carrier lifetime (τ_r), transit time (τ_t), and barrier height (V_b), were extracted, with V_b obtained from Mott–Schottky analysis. RW primarily affects transport, whereas recombination depends on both wave types, with PW influence more pronounced (see Table).

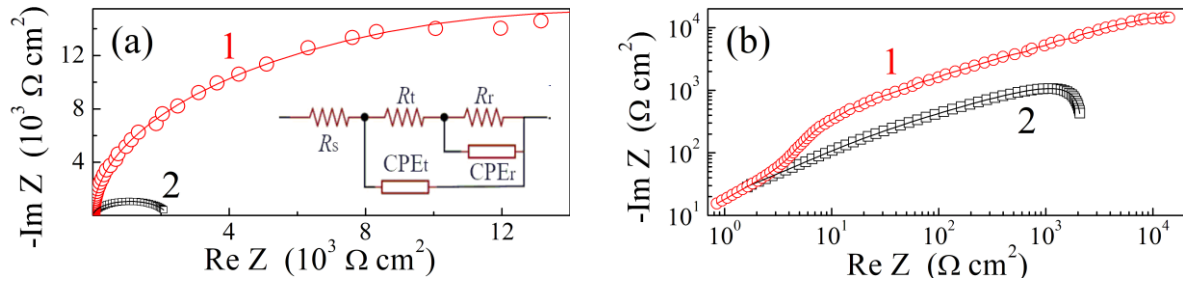


Fig. Impedance response of Si–PEDOT:PSS structures fabricated with (1, radial wave) and without ultrasound, in linear (a) and logarithmic (b) scales.

Points: experimental data; lines: fitted equivalent circuit (inset)

Table. Heterostructure parameters versus fabrication conditions

SC speed, rpm	3000			5000				
RTR time, min	20						60	
US wave	Non	PW	RW	Non	PW	RW	Non	PW
τ_t , 10^{-7} s	29±5	27±6	67±7	20±5	31±6	50±5	50±4	55±5
V_b , meV	820±10	810±3	790±10	830±25	810±10	770±10	670±25	750±30
τ_r , 10^{-5} s	17±3	6±1	13±3	18±3	1.7±0.2	5±2	11±3	2.2±0.6
n_r , 10^{-3}	973±3	985±3	970±4	973±5	990±5	971±5	989±4	993±4

1. Liu P. *et al.* Evolution of interfacial defects and energy losses during aging of organic photovoltaics. *Physica B Condens. Matter.* 2024. V. 677, 415707.